IMPLEMENTATION PLAN

Deep Learning–Based Neural Decoder for EEG Motor Imagery

A small hybrid CNN–Transformer that has to work for people it has never seen

TEAM

Ananya Gourishetty · Haarika D · Siddhanth Samal · Sathvik P

GUIDE

Dr. Mohan Allam

COURSE

CSE4006 Deep Learning · Group 5

School of Computer Science and Engineering, VIT-AP University

OVERVIEW

What we will cover

01

The problem

Why decoders that work for familiar people fail on someone new

02

The data

BCI Competition IV 2a: nine people, four imagined movements

03

What others have tried

Four published models, what each gives us, and the gap they leave

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Our design

HCT-Net, the preprocessing pipeline and Euclidean alignment

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How we test it

Leave-one-subject-out, the metrics, and six experiments

06

Plan

An 18-day build, the risks, and what we expect to learn

Accuracy collapses on a new person

A motor-imagery BCI turns imagined movements into commands, for example for someone who can no longer move their hands.

Today, every new user first sits through a long labelled calibration session. Skip it, and accuracy falls sharply.

Our goal: a decoder that works on a person it has never seen.

82.5% → **58.6%**

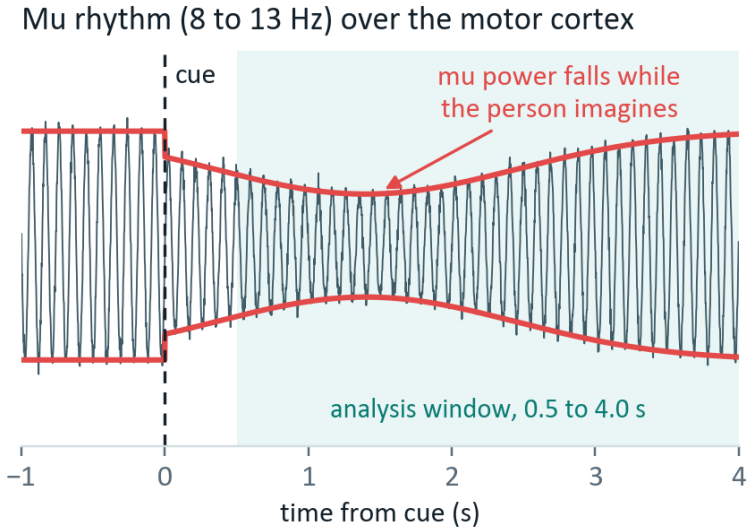
when the person was part of training

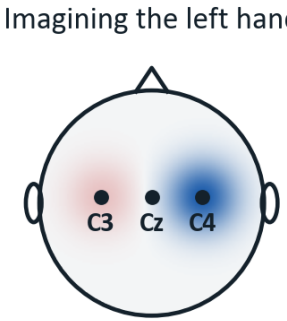
on a person it has never seen

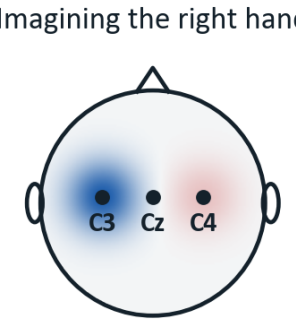
–23.9 points

CTNet on BCI IV-2a, four classes (Zhao et al., 2024). Chance is 25%.

What the decoder has to pick up







Imagine moving your left hand and the 8–13 Hz rhythm over the right side of your head gets weaker. Right hand, left side. The decoder has to find that drop inside very noisy signals.

µV

Tiny signal

A few millionths of a volt, mixed with eye, muscle and mains noise.

288

Little data

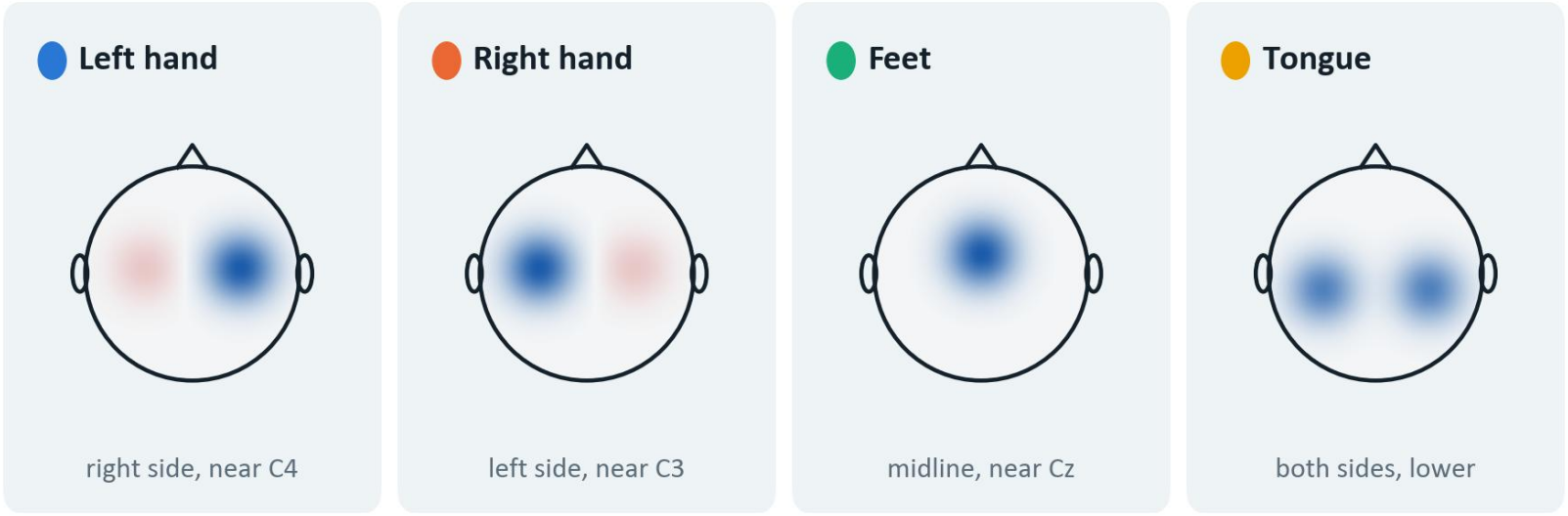
labelled trials per session, yet one trial is 22 × 875 numbers.

9

Every head differs

people, each with their own skull, cap fit and way of imagining.

Our data: 9 people, 4 imagined movements

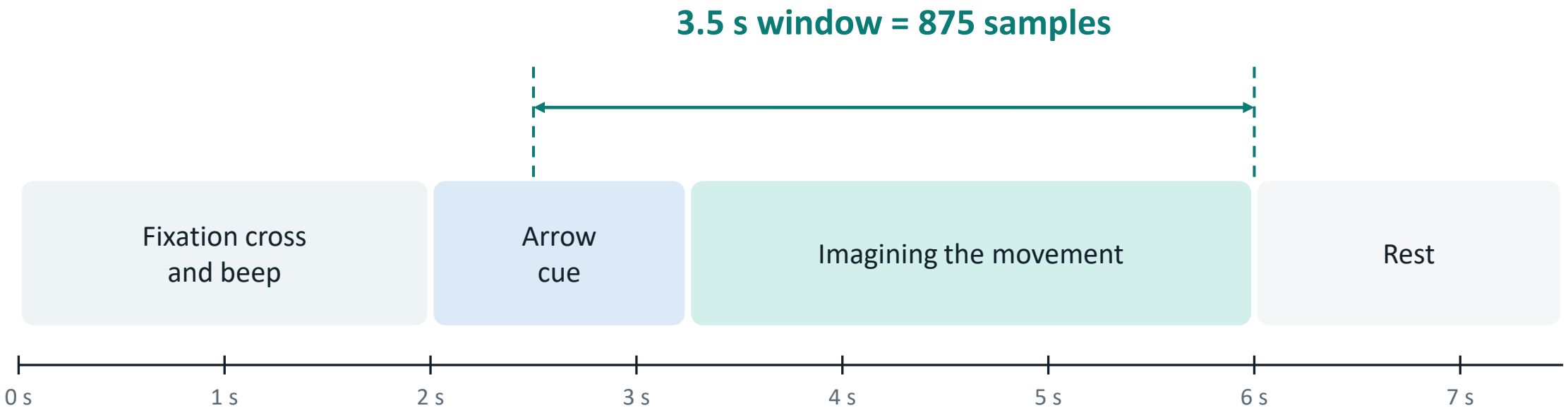


BCI Competition IV dataset 2a, recorded at Graz University of Technology. We chose it because all four papers we build on report results on it, so our re-runs can be checked against their published numbers.

Nine people gives nine leave-one-out folds: enough for a mean, a spread and a paired test, but too few to trust any single fold on its own.

9	people, A01 to A09
22	EEG electrodes; 3 eye channels dropped
250 Hz	sampling rate
2	sessions each, on different days
5,184	labelled trials in total
25%	chance level, as the four classes are balanced

We read 3.5 seconds of imagery per trial



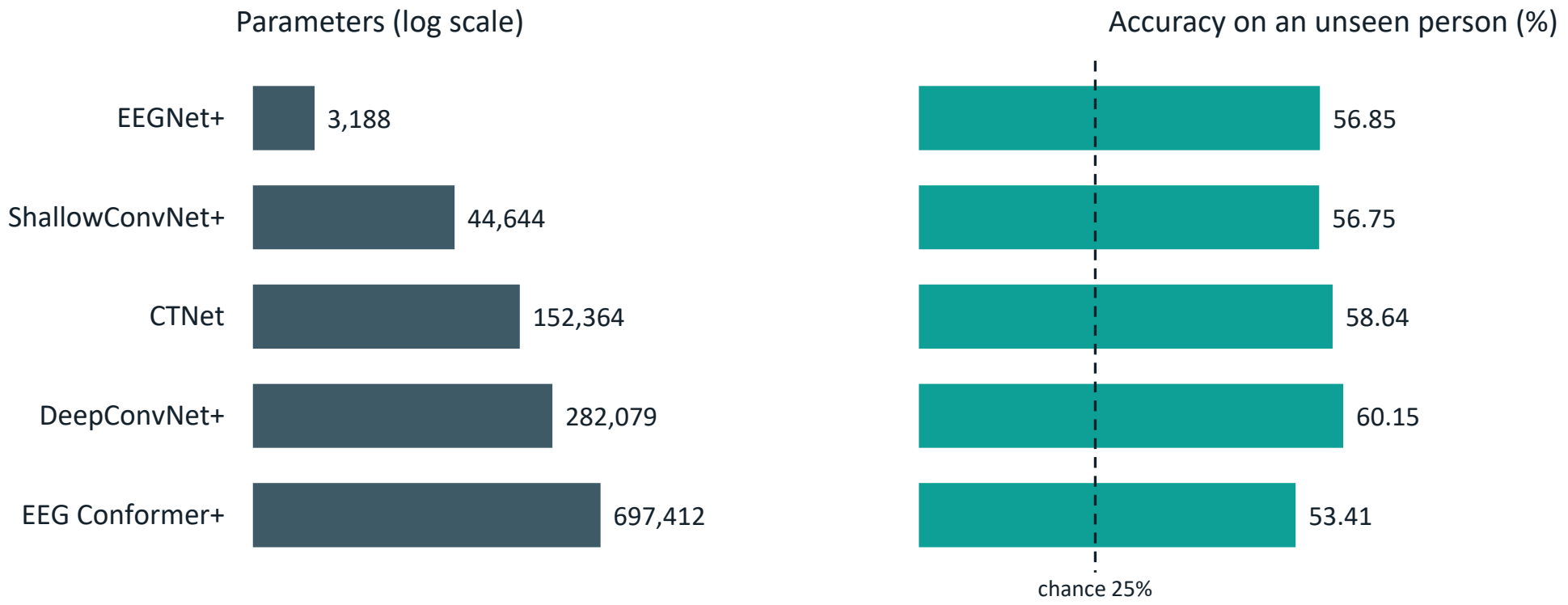
Why start 0.5 s after the arrow?

That first half-second is mostly the brain reacting to seeing the arrow. A model could learn that reaction and still know nothing about imagined movement.

Why stop at 6 s?

The imagery period ends there. The window gives 875 samples per electrode at 250 Hz, for every model alike.

Bigger models don’t fix the new-person problem



EEG Conformer has 219 times EEGNet’s parameters and scores 3.4 points lower. The best of the five is only 3.3 points ahead.

Leave-one-subject-out accuracy reported by Zhao et al. (2024); + marks models the CTNet authors re-ran under the same protocol. Parameter counts: braindecode reference models at our 875-sample input. Chance is 25%.

What we take from each paper

2018

EEGNet

Compact CNN

3,188 parameters

Learns frequency filters, then spatial filters, with very few weights.

WE TAKE

the whole convolutional front end

2023

ATCNet

CNN + attention + TCN

113,732 parameters

Applies attention inside sliding windows of the feature sequence.

WE TAKE

attention inside short windows

2023

EEG Conformer

CNN + Transformer

697,412 parameters

Convolution for local detail, then a Transformer across the trial.

WE TAKE

convolution first, attention after

2024

CTNet

CNN + Transformer

152,364 parameters

An EEGNet front end feeding a six-layer Transformer encoder.

WE TAKE

the encoder design, cut to two layers

None of the four is small, accurate and reliable on new people at the same time. That is the gap we work on.

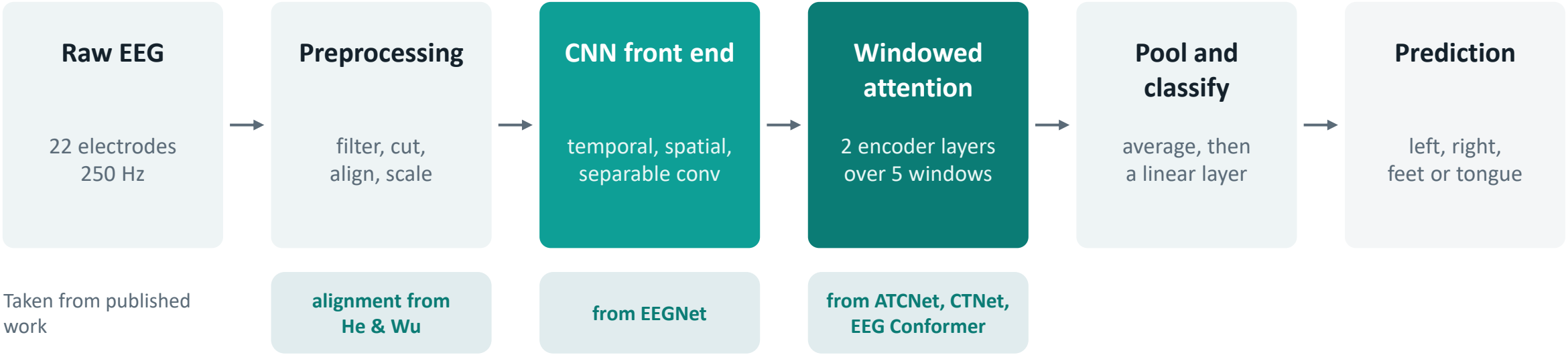
What we set out to do

1. Build one reproducible pipeline for BCI IV-2a, from raw files to results.
2. Re-run EEGNet, ATCNet, EEG Conformer, CTNet and FBCSP under one identical protocol.
3. Design HCT-Net: small, hybrid, and built for people it has never seen.
4. Test every model only on unseen people, with leave-one-subject-out.
5. Report every result, including the ones that go against us.

THE QUESTION BEHIND ALL FIVE

Can a model with about 21,000 parameters hold its own against much larger ones on a person it has never seen?

HCT-Net in one picture



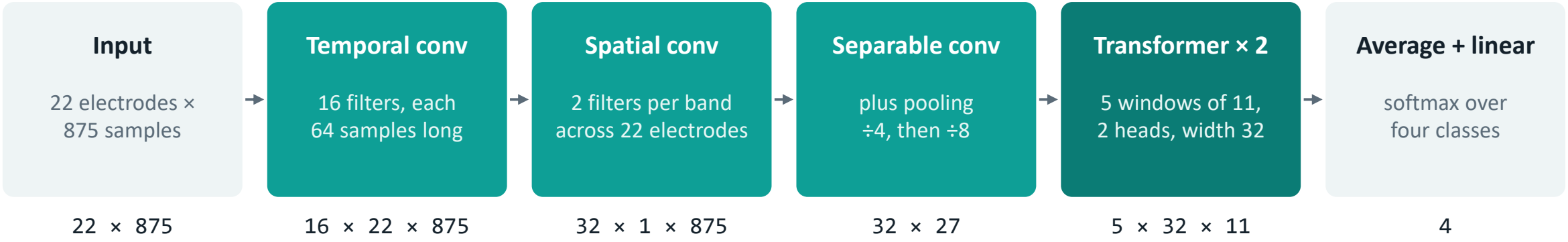
What is borrowed

Every block comes from published work: the front end from EEGNet, windowed attention from ATCNet, convolution then Transformer from EEG Conformer and CTNet, and alignment from He and Wu.

What is ours

Putting them together at this size for a 5,184-trial dataset, cutting attention to two layers, testing every model the same way, and measuring what each part adds.

Inside HCT-Net: 20,996 parameters



Where the 20,996 parameters go



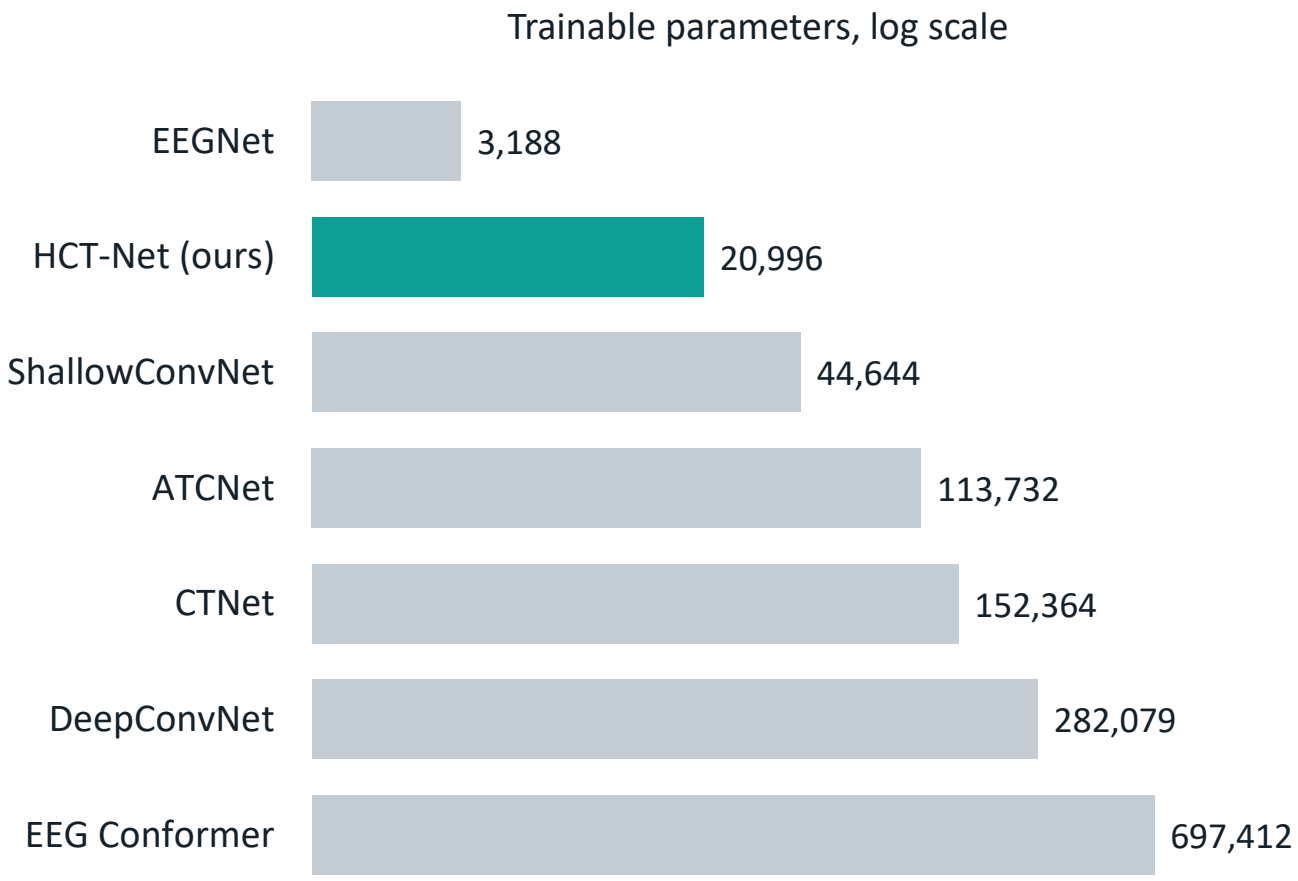
Why two encoder layers, not CTNet's six?

Six layers at this width would put 51,264 parameters in attention alone, more than our whole budget of 50,000.

Why keep it this small?

Each fold trains on only 4,032 trials. Every extra weight is another chance to memorise the seven training people instead of learning what they share.

Small, but not the smallest



1/5

the size of ATCNet

1/33

the size of EEG Conformer

6.6×

the size of EEGNet, our smallest baseline

Counts from the braindecode 1.8.1 reference implementations: 22 channels, 4 classes, 875-sample input.

One preprocessing pipeline for every model

1 Keep 22 channels

Drop the 3 eye channels after using them to spot artefacts.

2 Band-pass 4–38 Hz

Run forwards and backwards, so nothing shifts in time.

3 Cut the trial

0.5 to 4.0 s after the cue: 875 samples.

4 Reject artefacts

Any channel swinging more than 100 µV peak to peak.

5 Align each session

Euclidean alignment, using only unlabelled trials.

6 Standardise

Means and spreads from the training people only.

7 Augment

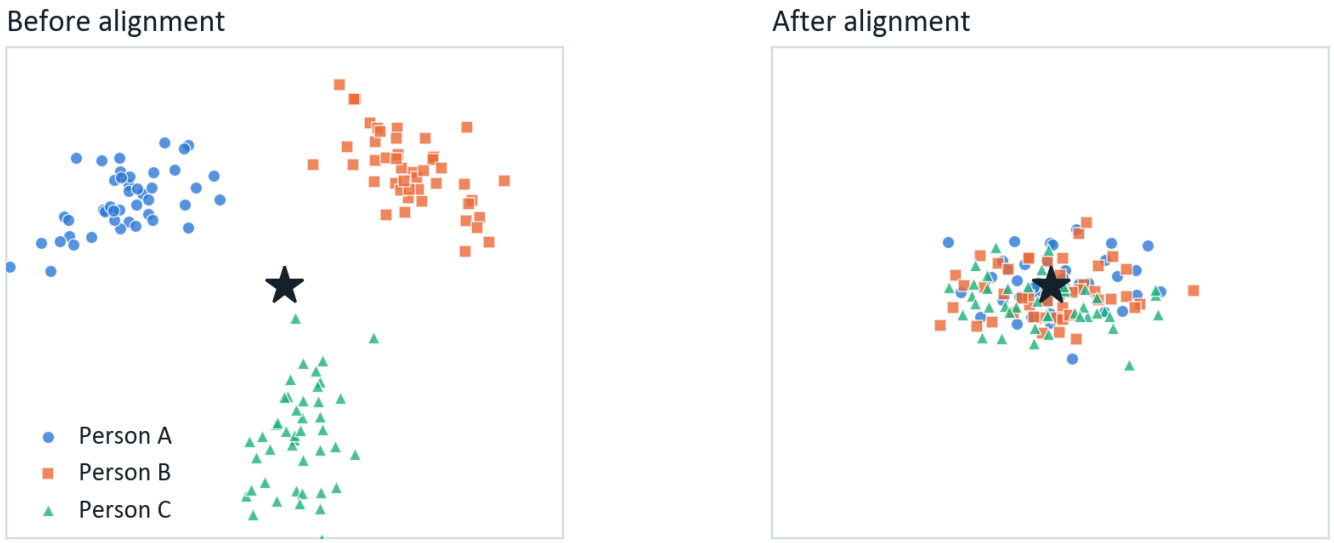
Stitch 8 segments from same-class trials. Training only.

8 Cache

Write once, reuse for every model and every fold.

The test person’s labels are never used. Their unlabelled trials go into one place only: their own alignment in step 5.

Alignment puts every person in one frame



Illustration, not measured data.

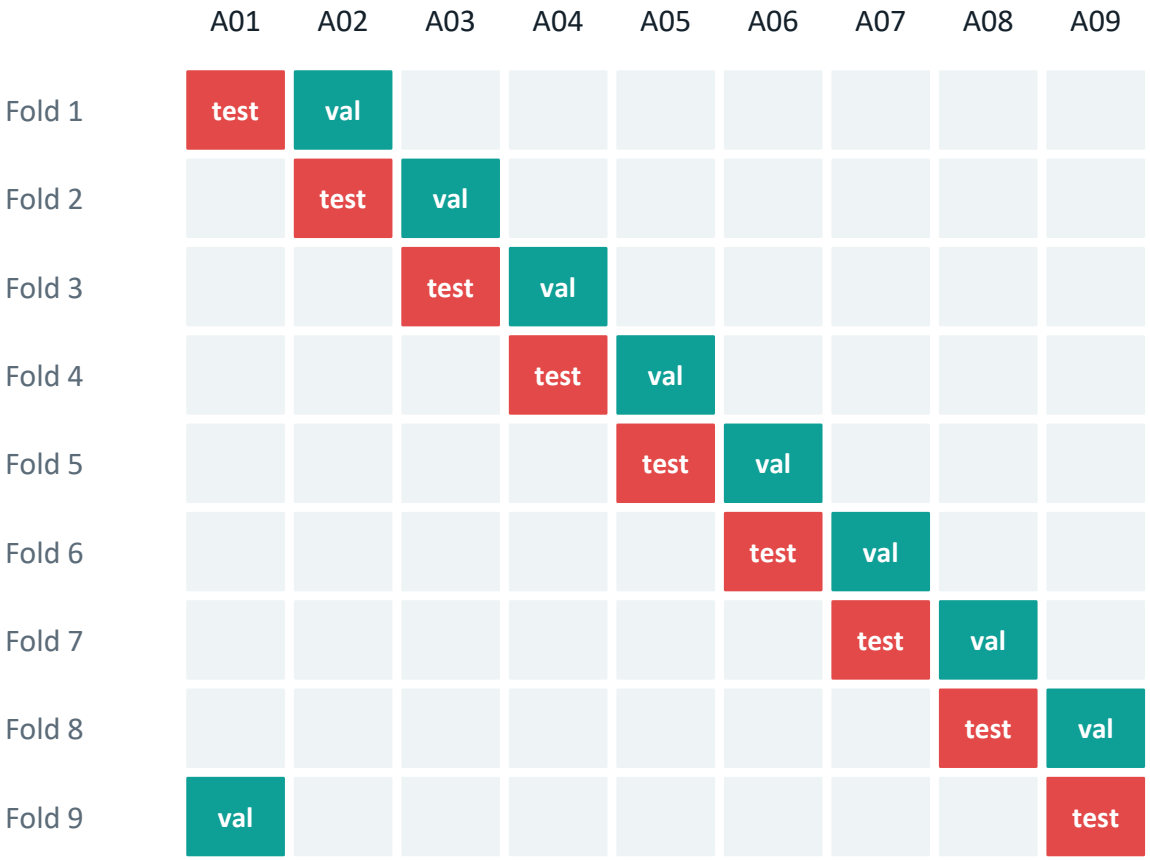
So a new user only has to wear the cap and record a few unlabelled trials. That is enough to align them before the model sees anything.

For each person

1. Average the covariance of all their trials. Call it R.
2. Multiply every trial by $R^{-1/2}$.
3. Now their average covariance is the identity, the star in the picture.

- ✓ **needs no labels**
- ✓ **adds zero parameters**
- ✓ **costs one 22 × 22 eigen-decomposition**

Every test is on a person the model never saw



In every fold, 7 people train, 1 validates and 1 is tested. The roles rotate.

What we measure

- Accuracy per person, mean and spread
- Cohen’s kappa, which corrects for chance
- Macro precision, recall and F1
- Confusion matrix, to catch feet vs tongue
- Size, training time and inference time

How we decide a difference is real

Wilcoxon signed-rank test over the nine people, Holm correction across the five comparisons, and three random seeds for every setting.

Six experiments, one question each

		runs
E1	**Baselines within subject** Do our versions match their papers?	45
E2	**Baselines on unseen people** How do they do under one protocol?	45
E3	**HCT-Net on unseen people** How does our model do?	9
E4	**Component study** Which parts actually matter?	45
E5	**Attention depth** Are two layers enough, or do six help?	9
E6	**Size and speed** Is HCT-Net really lighter?	—

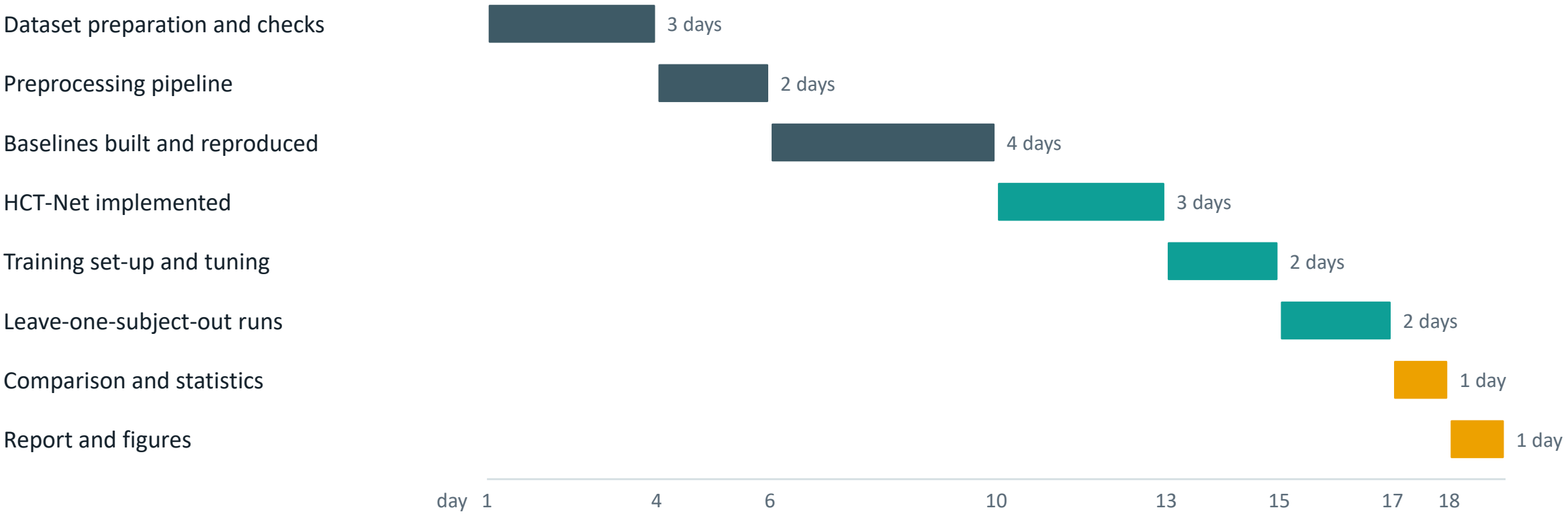
153 runs per seed, three seeds each. E6 is measured during E2 and E3.

E4: switch one part off at a time

Variant	Conv	Attention	Alignment	Augment
V0 full model	✓	✓	✓	✓
V1 no attention	✓	✗	✓	✓
V2 global attention	✓	global	✓	✓
V3 no alignment	✓	✓	✗	✓
V4 no augmentation	✓	✓	✓	✗
V5 convolution only	✓	✗	✗	✗

Every variant is trained and tested exactly like the full model, so a change in accuracy belongs to the part that was switched off.

An 18-day build, checked at every step



Nothing trains until the alignment check passes.

HCT-Net is written only after two baselines match their papers.

No leave-one-subject-out run starts until the folds are checked.

What could go wrong, and what we will do

RISK	WHY IT COULD HAPPEN	WHAT WE WILL DO
Too little data	5,184 trials in all; about 4,032 per training fold	**Augmentation, dropout, early stopping and a deliberately small model**
Baselines miss their papers	Each paper used its own preprocessing and protocol	**Use the authors' reference code and report both numbers side by side**
Nine people, wide spread	Some people produce only weak rhythms	**Per-person results, standard deviations and paired tests, not just means**
Not enough GPU time	153 runs per seed, three seeds each	**Cached data, mixed precision and a checkpoint every epoch**
HCT-Net does not win	The combination may not help on this dataset	**Report it plainly; the component study shows why**

What we expect, and what is new

WE EXPECT

not yet measured

- Accuracy on new people that holds up against much bigger models
- All of it at 20,996 parameters, about a fifth of ATCNet
- Less need for a long calibration session
- A clear answer either way, with the component study explaining it

WHAT IS NEW

- All four architectures compared under one leave-one-subject-out protocol
- A compact combination with attention sized for 5,184 trials
- A component study on unseen people, which the four papers do not report
- Accuracy weighed against model size under a single protocol

No single block is new. What we add is the combination, the fair protocol and the measurement.

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[9] Ang et al., “Filter bank common spatial pattern (FBCSP) in brain–computer interface,” Proc. IJCNN, 2008.

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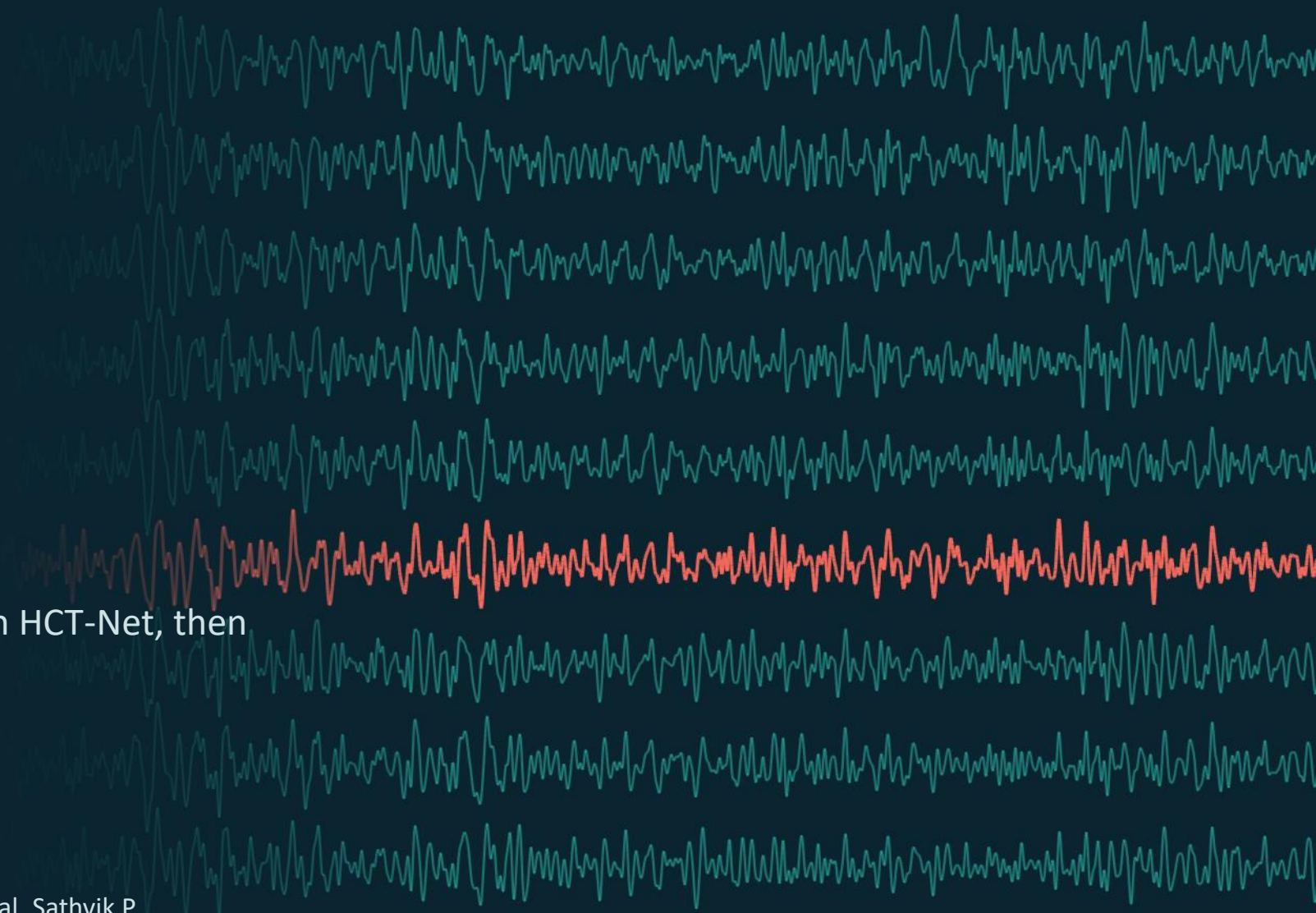
[14] Holm, “A simple sequentially rejective multiple test procedure,” Scandinavian Journal of Statistics, 1979.

[15] Wilcoxon, “Individual comparisons by ranking methods,” Biometrics Bulletin, 1945.

Thank you

Questions are welcome

Next: the pipeline and baselines first, then HCT-Net, then the leave-one-subject-out experiments.



Group 5 · Ananya Gourishetty, Haarika D, Siddhanth Samal, Sathvik P